

Badge Access: Forgot IN/OUT

Problem Statement

Given logs (name, time, IN/OUT), determine employees who missed an IN or OUT.

Variants

- Out-of-order logs; multiple days

- Mix

How to Leverage the Solution

Group per user, sort by time, and scan with a simple state machine.

Java Solution

Below is a concise reference implementation. Full comments omitted for brevity.

```
import java.util.*;

public final class BadgeAnomalyDetector {
    public static final class Event {
        public final String employee; public final int minutes; public final boolean isIn;
        public Event(String employee, int minutes, boolean isIn) { this.employee = employee; this.minutes
        = minutes; this.isIn = isIn; }
    }

    public static final class Result { public final Set<String> forgotIn = new HashSet<>(); public final
    al Set<String> forgotOut = new HashSet<>(); }

    public static Result findForgotInOut(List<Event> events) {
        Map<String, List<Event>> byEmployee = new HashMap<>();
        for (Event e : events) byEmployee.computeIfAbsent(e.employee, k -> new ArrayList<>()).add(e);
        Result result = new Result();
        for (Map.Entry<String, List<Event>> entry : byEmployee.entrySet()) {
            String name = entry.getKey();
            List<Event> list = entry.getValue();
            list.sort(Comparator.comparingInt(ev -> ev.minutes));
            boolean inside = false;
            for (Event e : list) {
                if (e.isIn) {
                    if (inside) result.forgotOut.add(name);
                    inside = true;
                } else {
                    if (!inside) result.forgotIn.add(name);
                    inside = false;
                }
            }
            if (inside) result.forgotOut.add(name);
        }
        return result;
    }

    public static int parseTimeToMinutes(String hhmm) {
        String[] p = hhmm.split(":");
        return Integer.parseInt(p[0]) * 60 + Integer.parseInt(p[1]);
    }
}
```